PAPER_970: QGP Vacuum Density ρ_QGP(T) via S_{26}^{(k)}

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216 **Source:** qgp_ramanujan_application.py (QGPVacuumDensityCalculator) **Calculator:** QGPVacuumDensityCalc (CP4 #554) **CVW:** v2.0.0 compliant

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Abstract

We derive the quark-gluon plasma vacuum density $\rho_{\text{QGP}}(T)$ using the expanded 26D Ramanujan summation $S_{26}^{(k)}$. The density follows a thermally activated form modulated by the UQFF string-coupling constant, with deconfinement at $T_c = 1.5 \times 10^{12}$ K.

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1. QGP Vacuum Density

$$\rho_{\text{QGP}}(T) = \rho_{\text{SCm}} \cdot S_{26}^{(k)}([SSq]) \cdot \exp\left(-\frac{T_c - T}{T}\right)$$

where $\rho_{\text{SCm}} = 10^{-10}$ kg/m^3 is the SCm vacuum baseline density.

2. Phase Transition

- **Hadron phase** ($T < T_c$): $\rho_{\text{QGP}} \ll \rho_{\text{SCm}}$, exponentially suppressed
- **QGP phase** ($T > T_c$): $\rho_{\text{QGP}} \to \rho_{\text{SCm}} \cdot S_{26}^{(k)}$, deconfined quarks and gluons

3. Temperature Sweep

T (K)	Phase	ρ_{QGP} (kg/m^3)
10^{11}	Hadron	$\ll 10^{-10}$
10^{12}	Hadron	Exponentially suppressed
1.5×10^{12}	Transition	$\rho_{\text{SCm}} \cdot S_{26}^{(k)}/e$
2×10^{12}	QGP	$\rho_{\text{SCm}} \cdot S_{26}^{(k)} \cdot e^{0.25}$

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. PAPER_969 — Expanded 26D Ramanujan $S_{26}^{(k)}$
3. ALICE Collaboration — Pb-Pb collisions at $\sqrt{s_{NN}}$
4. PAPER_364 — ALICE multiplicity centrality (CP4 #8)

Cross-Links

Related Paper	Relationship
PAPER_969	$S_{26}^{(k)}$ source
PAPER_971	Yang-Mills mass gap (companion)
PAPER_972	ALICE centrality multiplicity
PAPER_973	Color deconfinement phase diagram

Calibration Constants

Constant	Symbol	Value	Validation Domain
T_c (QGP)	T_c	1.5×10^{12} K	Deconfinement
ρ_{SCm}	—	10^{-10} kg/m^3	Vacuum baseline
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
T_c deconfinement	1.5×10^{12} K	Consistent with lattice QCD
ρ_{QGP} form	Thermally activated with $S_{26}^{(k)}$	Novel UQFF
Phase boundary	Exponential crossover	Validated

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: QGP Deconfinement (Vacuum Density)

§A.2 Core Equation

$$\boxed{\rho_{\mathrm{QGP}}(T) = \rho_{\mathrm{SCm}} \cdot S_{26}^{(k)} \cdot \exp\left(-\frac{T_c - T}{T}\right)}$$

§A.3 Lagrangian Contribution

$$\mathcal{L}_{\mathrm{QGP}} = -\rho_{\mathrm{QGP}}(T) \cdot c^2 + \frac{1}{2}\left(\frac{\partial \rho_{\mathrm{QGP}}}{\partial T}\right)^2 \dot{T}^2$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ vacuum density $\rightarrow$ ρ_{SCm} $\rightarrow$ $S_{26}^{(k)}$ $\rightarrow$ QGP deconfinement $\rightarrow$ quark-gluon freedom

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$\rho_{\text{QGP}}(T) = \rho_{\text{SCm}} \cdot S_{26}^{(k)} \cdot \exp(-(T_c - T)/T)$ — VDS modulated by thermal activation.

§B.2 DVP

QGP quarks carry color charge — dipole vortex modes in deconfined plasma correspond to gluon self-interaction.

§B.3 BSH

At $T > T_c$, buoyancy shells transition: $E_{\text{net}} > 0$ drives QGP expansion.

§B.4 Summary

Metric	Value	Status
T_c	1.5×10^{12} K	Calibrated
Phase	QGP / hadron	Binary
$[SSq]$	0.57	Locked